

Chapter-3: Calculus of VVFs and VFs

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Vector-Valued Functions and curves

- $\mathbf{r}(t) = \langle x(t), y(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j}$ is a vector-function in $\mathbb{R}^2$.

It traces a plane curve.

Example:

$\mathbf{r} = \langle a \cos t, a \sin t \rangle$ is a circle, plane curve, oriented counterclockwise.

- $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$ vector function in $\mathbb{R}^3$.

It traces a space curve.

Example:

$\mathbf{r} = \langle a \cos t, a \sin t, ct \rangle$ is a circular helix, a space curve.

- A line segment from point A to B is defined by the vector function

$$\mathbf{r}(t) = A + (B - A)t, \quad 0 \leq t \leq 1.$$

Example:

The line segment from the point $(1, -2, 0)$ to $(2, -1, 3)$:

$$\mathbf{r}(t) = (1, -2, 0) + (1, 1, 3)t, \quad 0 \leq t \leq 1.$$

- The part of the graph of $y = f(x)$ from $(a, f(a))$ to the point $(b, f(b))$ is a plane curve defined by

$$\mathbf{r}(t) = \langle t, f(t) \rangle, \quad a \leq t \leq b.$$

Example:

The part of the parabola $y = x^2$ from $(-1, 1)$ to $(2, 4)$ is

Derivatives and Integrals of Vector Value Functions

- If $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, then $\mathbf{r}'(t) = \langle x'(t), y'(t), z'(t) \rangle$

Differentiation Rules:

Suppose $\mathbf{u}$ and $\mathbf{v}$ are differentiable vector functions, c is a scalar, and f is a real-valued function. Then

- 1 $\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$
- 2 $\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$
- 3 $\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$
- 4 $\frac{d}{dt}[\mathbf{u}(t) \cdot \mathbf{v}(t)] = \mathbf{u}'(t) \cdot \mathbf{v}(t) + \mathbf{u}(t) \cdot \mathbf{v}'(t)$
- 5 $\frac{d}{dt}[\mathbf{u}(t) \times \mathbf{v}(t)] = \mathbf{u}'(t) \times \mathbf{v}(t) + \mathbf{u}(t) \times \mathbf{v}'(t)$
- 6 $\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t))$

- A curve $C: \mathbf{r} = \mathbf{r}(t)$ is **smooth** on $[a, b]$ if $\mathbf{r}'(t)$ is continuous and $\|\mathbf{r}'(t)\| \neq 0$ for all $t \in (a, b)$.
- $\mathbf{r}'(t)$ is **tangent** to the curve C at the point $\mathbf{r}(t)$

Integrals of Vector-valued functions:

Indefinite Integral:

$$\begin{aligned}
 \int \mathbf{r}(t) dt &= \int \langle x(t), y(t), z(t) \rangle dt \\
 &= \left\langle \int x(t) dt + k_1, \int y(t) dt + k_2, \int z(t) dt + k_3 \right\rangle \\
 &= \left\langle \int x(t) dt, \int y(t) dt, \int z(t) dt \right\rangle + \langle k_1, k_2, k_3 \rangle \\
 &= \mathbf{R}(t) + \mathbf{k},
 \end{aligned}$$

where $\mathbf{R}'(t) = \mathbf{r}(t)$ and $\mathbf{k} = k_1\mathbf{i} + k_2\mathbf{j} + k_3\mathbf{k}$

Definite integral:

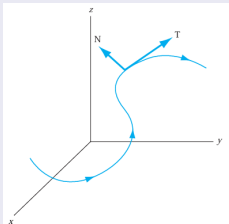
$$\begin{aligned}
 \int_a^b \mathbf{r}(t) dt &= \int_a^b \langle x(t), y(t), z(t) \rangle dt = \left\langle \int_a^b x(t) dt, \int_a^b y(t) dt, \int_a^b z(t) dt \right\rangle \\
 &= \mathbf{R}(b) - \mathbf{R}(a)
 \end{aligned}$$

Arclength, Unit Tangent and Unit Normal Vectors, Curvature

- Let $C : \mathbf{r} = \mathbf{r}(t)$, $a \leq t \leq b$ be a **smooth curve**.

Arc-length: $L = \int_a^b \|\mathbf{r}'(t)\| dt$

- $s(t) = \int_a^t \|\mathbf{r}'(u)\| du$, $a < t < b$ is **arc-length function**
- Differential of arc-length function: $ds = \|\mathbf{r}'\| dt$.



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Unit Tangent Vector: $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$

- $\mathbf{T}$ points in the direction of the increasing parameter t .

Unit Normal Vector: $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|}$, $\|\mathbf{T}'(t)\| \neq 0$.

- $\mathbf{N}$ is undefined if $\|\mathbf{T}'(t)\| = 0$
- $\mathbf{N}$ points to the concave side of C
- $\mathbf{N} \perp \mathbf{T} \Rightarrow \mathbf{T} \cdot \mathbf{N} = 0$.

Curvature: $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|}$

- The more a curve bends at a point, the faster the unit tangent vector is changing direction there.
- κ is a scalar and $\kappa \geq 0$.
- $\kappa = 0 \Rightarrow \mathbf{T}$ does not change. That is C is a straight line.
- κ is very large means C is very bending.
- κ close to zero means C is nearly a straight line.
- curvature of a circle of radius a is $\frac{1}{a}$.

Radius of curvature: $r(t) = \frac{1}{\kappa(t)}$.

Example:

Give the curve $C: \mathbf{r}(t) = \langle e^t \cos t, e^t \sin t, e^t \rangle$, $0 \leq t \leq 2\pi$. Find

- a) L b) $\mathbf{T}$ c) $\mathbf{N}$ d) κ and radius of curvature

Solution:

- a) $L = \int_0^{2\pi} \|\mathbf{r}'(t)\| dt = \int_0^{2\pi} \sqrt{3}e^t dt = \sqrt{3} [e^{2\pi} - 1]$
- b) $\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|} = \left\langle \frac{\cos t - \sin t}{\sqrt{3}}, \frac{\sin t + \cos t}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right\rangle$
 $\mathbf{T}'(t) = \left\langle \frac{-\cos t - \sin t}{\sqrt{3}}, \frac{\cos t - \sin t}{\sqrt{3}}, 0 \right\rangle$ and $\|\mathbf{T}'(t)\| = \frac{\sqrt{6}}{3}$
- c) $\mathbf{N}(t) = \frac{\mathbf{T}'(t)}{\|\mathbf{T}'(t)\|} = \left\langle \frac{-\cos t - \sin t}{\sqrt{2}}, \frac{\cos t - \sin t}{\sqrt{2}}, 0 \right\rangle$
- d) $\kappa(t) = \frac{\|\mathbf{T}'(t)\|}{\|\mathbf{r}'(t)\|} = \frac{\frac{\sqrt{6}}{3}}{\sqrt{3}e^t} = \frac{\sqrt{2}}{3}e^{-t}$ and $r(t) = \frac{1}{\kappa(t)} = \frac{3e^t}{\sqrt{2}}$

Scalar and Vector Fields

- Functions of two or more variables are called **Scalar-Fields**.
- **Vector-fields** are vector functions whose components are scalar fields.

Example:

- Velocity vector fields
- Gradient vector-field
- Force vector-field

A vector field $\mathbf{F}$ on $\mathbb{R}^3$ is defined by

$$\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$$

where M, N, P are **scalar fields** in $\mathbb{R}^3$

Example

$\mathbf{F}(x, y, z) = \langle x + yz^2, 3y - xz, xye^z \rangle$, is a vector field in $\mathbb{R}^3$.

Navigation icons: back, forward, search, etc.

- A vector field $\mathbf{F}$ on $\mathbb{R}^2$ is defined by

$$\mathbf{F}(x, y) = \langle M(x, y), N(x, y) \rangle = M(x, y)\mathbf{i} + N(x, y)\mathbf{j}$$

- The best way to picture a vector field is to draw the arrow representing the vector $\mathbf{F}(x, y)$ starting at the point (x, y) .
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Example

For the vector field $\mathbf{F}(x, y) = \langle x + y, 3y - x \rangle$, evaluate

a) $\mathbf{F}(1, 0)$

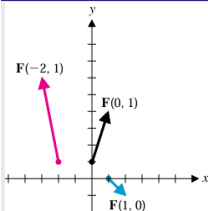
b) $\mathbf{F}(0, 1)$

a) $\mathbf{F}(-2, 1)$

and plot each vector (x, y) using the point (x, y) as the initial point.

Navigation icons: back, forward, search, etc.

Solution:



- a) $\mathbf{F}(1, 0) = \langle 1, -1 \rangle$. Plot $\langle 1, -1 \rangle$ with initial point at $(1, 0)$, and terminal point at $(2, -1)$.
- b) $\mathbf{F}(0, 1) = \langle 1, 3 \rangle$. Plot $\langle 1, 3 \rangle$, taking $(0, 1)$ as its initial point, and terminal point $(1, 4)$.
- c) $\mathbf{F}(-2, 1) = \langle -1, 5 \rangle$. Plot $\langle -1, 5 \rangle$ by placing its initial point at $(-2, 1)$ and its terminal point at $(-1, 5)$.

Definitions:

- $\mathbf{F}$ is **conservative** if $\mathbf{f} = \nabla f$ for some scalar field f .
- f is called a potential for $\mathbf{F}$.
- If f is a potential, so is $f + k$, for any constant k .

Example:

$\mathbf{F} = \langle y^2 - 1, 2xy, 4z^3 \rangle$ is **conservative** because there is a function $f(x, y, z) = xy^2 - x + z^4$ such that $\nabla f = \langle y^2 - 1, 2xy, 4z^3 \rangle = \mathbf{F}$

Divergence and Curl

Definition

- Let $\mathbf{F}(x, y, z) = M(x, y, z)\mathbf{i} + N(x, y, z)\mathbf{j} + P(x, y, z)\mathbf{k}$.
 - Divergence of $\mathbf{F}$ is the scalar field $\operatorname{div} \mathbf{F} = M_x + N_y + P_z$.
 - Curl of $\mathbf{F}$ is the vector field

$$\operatorname{curl} \mathbf{F} = (P_y - N_z)\mathbf{i} + (M_z - P_x)\mathbf{j} + (N_x - M_y)\mathbf{k}$$
 - $\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$ del operator-differential operator

- The product of ∇ and f is the gradient of f :

$$\nabla f = \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k} \right) = \frac{\partial f}{\partial x}\mathbf{i} + \frac{\partial f}{\partial y}\mathbf{j} + \frac{\partial f}{\partial z}\mathbf{k}.$$

- The dot product of ∇ and $\mathbf{F}$ is the divergence of $\mathbf{F}$:

$$\begin{aligned}\nabla \cdot \mathbf{F} &= \left(\frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k} \right) \cdot (M\mathbf{i} + N\mathbf{j} + P\mathbf{k}) \\ &= \frac{\partial M}{\partial x} + \frac{\partial N}{\partial y} + \frac{\partial P}{\partial z} = \operatorname{div} \mathbf{F}\end{aligned}$$

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- The cross product of ∇ with $\mathbf{F}$ is the curl of $\mathbf{F}$:

$$\nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ M & N & P \end{vmatrix} = \operatorname{curl} \mathbf{F}.$$

- There are two relationships between gradient, divergence and curl that are fundamental to vector analysis:

- the **curl** of a **gradient** is the **zero vector**, $\nabla \times \nabla f = \mathbf{0}$.
- the **divergence** of a **curl** is (the number) **zero**, $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.

Example:

Find the divergence and curl of $\mathbf{F}(x, y, z) = xz\mathbf{i} + xyz\mathbf{j} - y^2\mathbf{k}$.

Solution

- $\operatorname{div} \mathbf{F} = M_x + N_y + P_z = z + xz$
- $\operatorname{curl} \mathbf{F} = \nabla \times \mathbf{F} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xz & xyz & -y^2 \end{vmatrix} = \langle -2y - xy, x, yz \rangle$

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Test for Conservative Vector-Field:

- Let $\mathbf{F} = M\mathbf{i} + N\mathbf{j} + P\mathbf{k}$ be a vector field whose component functions have continuous partial derivatives. Then $\mathbf{F}$ is a conservative if and only if

$$\frac{\partial P}{\partial y} = \frac{\partial N}{\partial z}, \quad \frac{\partial M}{\partial z} = \frac{\partial P}{\partial x}, \quad \frac{\partial N}{\partial x} = \frac{\partial M}{\partial y}$$

Example:

Show that $\mathbf{F} = \langle e^x \cos y + yz, xz - e^x \sin y, xy - z \rangle$ is conservative and find a potential function for it

Solution:

- Let $M = e^x \cos y + yz$, $N = xz - e^x \sin y$, $P = xy - z$.
- $P_y = x = N_z$, $M_z = y = P_x$, $N_x = -e^x \sin y = M_y$.
- $\mathbf{F}$ is conservative.

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Solution:

- Let f a potential of $\mathbf{F}$. Then $\nabla f = \mathbf{F}$

$$\begin{cases} f_x = e^x \cos y + yz, \\ f_y = xz - e^x \sin y, \\ f_z = xy - z \end{cases} \quad (1)$$

- Integrating (1) with respect to x , y , z , respectively, we get

$$\begin{cases} f = e^x \cos y + xyz + g(x, y), \\ f = xyz + e^x \cos y + h(x, z), \\ f = xyz - \frac{z^2}{2} + k(x, y) \end{cases} \quad (2)$$

- From (2), $f(x, y, z) = e^x \cos y + xyz - \frac{z^2}{2}$, which is a potential function for $\mathbf{F}$.

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Line integrals of Scalar Functions

Definition

Let f be a scalar field on defined on a smooth curve

$C : \mathbf{r} = \mathbf{r}(t)$, $a \leq t \leq b$. Then the **line integral** of f along C is defined by

$$\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt \quad (3)$$

- For $f = f(x, y)$ and $C : \mathbf{r}(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$,

$$\int_C f ds = \int_a^b f(x(t), y(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} dt$$

- For $f = f(x, y, z)$ and $C : \mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$,

$$\int_C f ds = \int_a^b f(x(t), y(t), z(t)) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2} dt$$

Steps to evaluate $\int_C f ds$:

- Represent C by a vector function $\mathbf{r}(t)$ for $a \leq t \leq b$ if not.
- Find $f(\mathbf{r}(t))$ (value of f along C).
- Find $\mathbf{r}'(t)$ and then $ds = \|\mathbf{r}'(t)\| dt$
- Evaluate the line integral: $\int_C f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$

Properties of $\int_C f ds$:

- If $C = C_1 \cup \dots \cup C_n$, where $C_1, \dots, C_n$ are smooth curves and the initial point of C_{i+1} is the terminal point of C_i . Then

$$\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds + \dots + \int_{C_n} f ds$$
- $\int_C f ds = \int_{-C} f ds$, $-C$ denotes the curve consisting of the same points as C but with the opposite orientation.

- For any constant k , $\int_C k f ds = k \int_C f ds$.
- For $f = 1$, $\int_C f ds = \int_C ds$ gives the arc length of the curve C .

Examples

- a) If arc-length of C is 7 and $f = 3$ then $\int_C f ds = \dots\dots$
- b) If arc-length of C is 2π and $f = -\sqrt{6}$ then $\int_C f ds = \dots\dots$

Example:

Evaluate $\int_C 2y\sqrt{1+x^2}ds$, C consists of the parabola $y = \frac{x^2}{2}$ from $(0,0)$ to $(1, \frac{1}{2})$ and the line segment from $(1, \frac{1}{2})$ to $(1, \frac{3}{2})$.

Solution

- $C = C_1 \cup C_2$, $C_1 : \mathbf{r}_1(t) = \left\langle t, \frac{t^2}{2} \right\rangle$, $0 \leq t \leq 1$ and $C_2 : \mathbf{r}_2(t) = \left\langle 1, \frac{1}{2} + t \right\rangle$, $0 \leq t \leq 1$.
- $f(\mathbf{r}_1(t)) = t^2\sqrt{1+t^2}$ and $f(\mathbf{r}_2(t)) = \sqrt{2}(2t+1)$.
- $\mathbf{r}'_1(t) = \langle 1, t \rangle$ and $\mathbf{r}'_2(t) = \langle 0, 1 \rangle$
 $f(\mathbf{r}_1(t))\|\mathbf{r}'_1(t)\| = t^2 + t^4$ and $f(\mathbf{r}_2(t))\|\mathbf{r}'_2(t)\| = \sqrt{2}(2t+1)$.
- $\int_{C_1} f ds = \int_0^1 (t^2 + t^4) dt = \left(\frac{x^3}{3} + \frac{x^5}{5} \right) \Big|_0^1 = \frac{8}{15}$ and $\int_{C_2} f ds = \int_0^1 \sqrt{2}(2t+1) dt = \sqrt{2} \left(t^2 + t \right) \Big|_0^1 = 2\sqrt{2}$
- $\int_C f ds = \int_{C_1} f ds + \int_{C_2} f ds = \frac{8+30\sqrt{2}}{15}$

Example:

Find the mass and center of mass of a wire in the shape of the helix $x = t$, $y = \cos t$, $z = \sin t$, $0 \leq t \leq 2\pi$ if the density at any point is equal to the square of the distance from the origin.

Solution

- $\rho = x^2 + y^2 + z^2 = t^2 + \cos^2 t + \sin^2 t = t^2 + 1$.
- $ds = \sqrt{(1)^2 + (-\sin t)^2 + (\cos t)^2} dt = \sqrt{2} dt$
- $m = \int_C \rho ds = \int_0^{2\pi} (t^2 + 1) \sqrt{2} dt = \sqrt{2} \left(\frac{t^3}{3} + t \right) \Big|_0^{2\pi} = \frac{\sqrt{8}(4\pi^2 + 3)\pi}{3}$
- Let $(\bar{x}, \bar{y}, \bar{z})$ the center of mass of the wire, where

$$\begin{aligned} \bar{x} &= \frac{1}{m} \int_C x \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^3 + t) \sqrt{2} dt \\ &= \frac{3}{2\pi(4\pi^2 + 3)} \left(\frac{t^4}{4} + \frac{t^2}{2} \right) \Big|_0^{2\pi} = \frac{6\pi^3 + 3}{4\pi^2 + 3} \end{aligned}$$

$$\begin{aligned} \bar{y} &= \frac{1}{m} \int_C y \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^2 + 1) \cos t \sqrt{2} dt \\ &= \frac{3}{2\pi(4\pi^2 + 3)} (t^2 \sin t + 2t \cos t - \sin t) \Big|_0^{2\pi} = \frac{6}{4\pi^2 + 3} \\ \bar{z} &= \frac{1}{m} \int_C z \rho ds = \frac{3}{\sqrt{8}(4\pi^2 + 3)\pi} \int_0^{2\pi} (t^2 + 1) \sin t \sqrt{2} dt \\ &= \frac{3}{2\pi(4\pi^2 + 3)} (-t^2 \cos t + 2t \sin t + \cos t) \Big|_0^{2\pi} = \frac{-6\pi}{4\pi^2 + 3} \end{aligned}$$

Line integral of a Vector-Field

Definition:

Let $\mathbf{F}$ be a continuous vector field defined on a smooth curve $C : \mathbf{r} = \mathbf{r}(t)$, $a \leq t \leq b$. Then the line integral of $\mathbf{F}$ along C is

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \mathbf{r}'(t) dt \quad (4)$$

- For $\mathbf{F} = \langle M, N \rangle$ and $C : \mathbf{r}(t) = \langle x(t), y(t) \rangle$, $a \leq t \leq b$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b [\mathbf{F}(x(t), y(t)) \cdot \langle x'(t), y'(t) \rangle] dt$$

- For $\mathbf{F} = \langle M, N, P \rangle$ and $C : \mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, $a \leq t \leq b$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b [\mathbf{F}(x(t), y(t), z(t)) \cdot \langle x'(t), y'(t), z'(t) \rangle] dt$$

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- If $\mathbf{F}$ is a **force vector field**, the work done by $\mathbf{F}$ in moving a particle along the curve C can be written simply as

$$W = \int_C \mathbf{F}(x, y, z) \cdot d\mathbf{r}$$

Example

Compute the work done by the force field $\mathbf{F} = \langle 4y, 2xz, 3y \rangle$ acting on an object as it moves along the helix defined by $x = 2 \cos t$, $y = 2 \sin t$, $z = 3t$, from the point $(2, 0, 0)$ to $(-2, 0, 3\pi)$.

Solution:

- $W = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_C 4y dx + 2xz dy + 3y dz$.
- The point $(2, 0, 0)$ corresponds to $t = 0$ and $(-2, 0, 3\pi)$ corresponds to $t = \pi$.

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- $dx = -2 \sin t \, dt$, $dy = 2 \cos t \, dt$ and $dz = 3 \, dt$.

$$\begin{aligned} W &= \int_0^\pi [-16 \sin^2 t + 24t \cos^2 t + 18 \sin t] \, dt \\ &= \int_0^\pi [-8 + 8 \cos 2t + 12t + 12t \cos 2t + 18 \sin t] \, dt \\ &= 36 - 8\pi + 6\pi^2 \end{aligned}$$

Flow and Circulation:

- If $\mathbf{F}$ is a **velocity vector-field** of a fluid flowing through a region in space, then the flow along the curve C is

$$\text{flow} = \int_C \mathbf{F} \cdot d\mathbf{r}$$

- If C is a closed, the flow is called the **circulation** around the curve C .

Steps to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$:

- Represent C as a vector function $\mathbf{r}(t)$, $a \leq t \leq b$, if not.
- Find $\mathbf{F}(\mathbf{r}(t))$, $\mathbf{r}'(t)$ and $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)$.
- Evaluate $\int_a^b [\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)] \, dt$

Properties of $\int_C \mathbf{F} \cdot d\mathbf{r}$:

- Reversing the orientation of C reverses the sign of the integral; that is,

$$\int_{-C} \mathbf{F} \cdot d\mathbf{r} = - \int_C \mathbf{F} \cdot d\mathbf{r}$$

- If C is made by joining a finite number $C_1, \dots, C_n$ of curves end to end, then $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_{C_1} \mathbf{F} \cdot d\mathbf{r} + \dots + \int_{C_n} \mathbf{F} \cdot d\mathbf{r}$

Fundamental Theorem of Line Integrals

- The Fundamental Theorem of Calculus:

$$\int_a^b F'(x) dx = F(b) - F(a)$$

where $F'(x)$ is continuous on $[a, b]$.

Theorem: The Fundamental Theorem of Line Integrals

Let $C : \mathbf{r} = \mathbf{r}(t)$, $a \leq t \leq b$ be a smooth curve. Let f be a differentiable function of two or three variables whose gradient vector ∇f is continuous on C and $\mathbf{F} = \nabla f$. Then

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

- We can evaluate the line integral of a conservative vector field (the gradient vector field of the potential function) simply by knowing the value of f at the endpoints of C .

Example: Evaluate the line integral

- a) $\int_C 2x dx - y^2 dy - \frac{4}{1+z^2} dz$, C is a path from the origin to $(3, 3, 1)$.
 b) $\int_C e^y dx + x e^y dy + (1+z) e^z dz$, $C : \mathbf{r}(t) = \langle t, t^2, t^3 \rangle$, $0 \leq t \leq 1$.

Solution:

- a) $\mathbf{F} = \left\langle 2x, -y^2, -\frac{4}{1+z^2} \right\rangle$ is conservative with potential

$$f = x^2 - \frac{y^3}{3} - 4 \tan^{-1} z.$$

$$\int_C 2x dx - y^2 dy - \frac{4}{1+z^2} dz = \left(x^2 - \frac{y^3}{3} - 4 \tan^{-1} z \right) \Big|_{(0,0,0)}^{(3,3,1)} = -\pi$$

- b) $\mathbf{F} = \langle e^y, x e^y, (1+z) e^z \rangle$ is conservative with potential $f = x e^y + z e^z$.

$$\mathbf{r}(0) = (0, 0, 0) \text{ and } \mathbf{r}(1) = (1, 1, 1).$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(1)) - f(\mathbf{r}(0)) = 2e$$

Line integrals of conservative vector fields are independent of path.

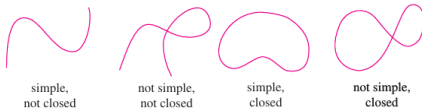
- $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path in a region D if and only if $\int_C \mathbf{F} \cdot d\mathbf{r} = 0$ for every closed path in D .

Suppose that $\mathbf{F}(x, y, z) = \langle M(x, y, z), N(x, y, z), P(x, y, z) \rangle$ is a vector field whose components M , N and P have continuous first partial derivatives in $\mathbb{R}^3$. Then the following are equivalent:

- 1 $\mathbf{F}$ is conservative.
- 2 $\int_C \mathbf{F} \cdot d\mathbf{r}$ is independent of path.
- 3 $\int_C \mathbf{F} \cdot d\mathbf{r}$ for every piecewise-smooth closed curve C .
- 4 $\mathbf{F}$ is a gradient field ($\mathbf{F} = \nabla f$ for some potential function f).

Green's Theorem:

- A curve is called closed if its terminal point coincides with its initial point, that is, $\mathbf{r}(a) = \mathbf{r}(b)$.
- A **simple curve** is a curve that doesn't intersect itself anywhere between its endpoints.

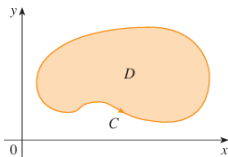


- Green's Theorem gives the relationship between a line integral around a simple closed curve C and a double integral over the plane region D bounded by C .
- Assume that D consists of all points inside C and all points on C .
- The **positive orientation** of a simple closed curve C refers to a single counterclockwise traversal of C .

Theorem: Green's Theorem

Let C be a simple closed positively oriented path in the plane. Let D be the region bounded by C . Let M , N , $\frac{\partial M}{\partial y}$ and $\frac{\partial N}{\partial x}$ be continuous on an open region that contains D , then

$$\int_C M dx + N dy = \iint_D (N_x - M_y) dA$$



Example

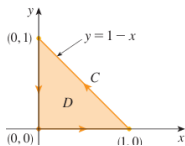
Compute the work done by $\mathbf{F}(x, y) = (y - x^2 e^x)\mathbf{i} + (\cos(2y^2) - x)\mathbf{j}$ in moving a particle counter clockwise about the rectangular path C having vertices $(0, 1)$, $(1, 1)$, $(1, 3)$ and $(0, 3)$.

Solution

$$\begin{aligned} \text{Work} &= \oint_C \mathbf{f} \cdot d\mathbf{r} = \iint_D \left(\frac{\partial}{\partial x} [\cos(2y^2) - x] - \frac{\partial}{\partial y} [y - x^2 e^x] \right) dA \\ &= \iint_D -2 dA = -2 \text{ Area}(D) = -4. \end{aligned}$$

Example

Evaluate $\int_C x^4 dx + xy dy$, where C is the triangular curve consisting of the line segments from $(0, 0)$ to $(1, 0)$, from $(1, 0)$ to $(0, 1)$, and from $(1, 0)$ to $(0, 0)$.



Solution:

C is piecewise-smooth, positively oriented, simple closed curve and $M = x^4$, $N = xy$ have continuous partial derivatives Right column
 $D = \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq 1 - x\}$

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= \iint_D (N_x - M_y) dA = \int_0^1 \int_0^{1-x} y dx dy = \frac{1}{2} \int_0^1 (1-x)^2 dx \\ &= -\frac{1}{6}(1-x)^3 \Big|_0^1 = \frac{1}{6}\end{aligned}$$

Example

Evaluate $\oint_C (3y - e^{\sin x}) dx + (7x + \sqrt{1+y^4}) dy$, where C is the circle $x^2 + y^2 = 9$.

Solution

- Let $M = 3y - e^{\sin x} \Rightarrow M_y = 3$, $N = 7x + \sqrt{1+y^4} \Rightarrow N_x = 7$
- $N_x - M_y = 7 - 3 = 4$ and $D = [0, 3] \times [0, 2\pi]$
- $\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (N_x - M_y) dA = \iint_D 4 dA = 4 \text{ Area}(D) = 36\pi$.

Example

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle \sqrt{x} + y^3, x^2 + \sqrt{y} \rangle$ and C consists of the arc of the curve $y = \sin x$ from $(0, 0)$ to $(\pi, 0)$ and the line segment from $(\pi, 0)$ to $(0, 0)$.

Solution

■ C is piecewise-smooth, negatively oriented, simply closed curve.

■ $M_y = 3y^2$, $N_x = 2x \Rightarrow N_x - M_y = 2x - 3y^2$

■ $D = \{(x, y) | 0 \leq x \leq \pi, 0 \leq y \leq \sin x\}$

$$\begin{aligned}\int_C \mathbf{F} \cdot d\mathbf{r} &= -\iint_D (N_x - M_y) dA = \int_0^\pi \int_0^{\sin x} (-2x + 3y^2) dy dx \\&= \int_0^\pi \left[-2xy + y^3 \right]_0^{\sin x} dx = \int_0^\pi \left[-2x \sin x + \sin^3 x \right] dx \\&= \left[2x \cos x + 2 \sin x - \frac{1}{3} \sin^2 x \cos x - \frac{2}{3} \cos x \right]_0^\pi \\&= \frac{4}{3} - 2\pi\end{aligned}$$

Example

A particle starts at the point $(-2, 0)$, moves along the x -axis to $(2, 0)$, and then along the semicircle $y = \sqrt{4 - x^2}$ to the starting point. Find the work done on this particle by the force field $\mathbf{F} = \langle x, x^3 + 3xy^2 \rangle$.

Solution

■ $M_y = 0$, $N_x = 3y^2 \Rightarrow N_x - M_y = 3y^2 = 3r^2 \sin^2 \theta$.

■ $D = [0, 2] \times [0, \pi]$ and $dA = r dr d\theta$

$$\begin{aligned}W &= \int_C \mathbf{F} \cdot d\mathbf{r} = \iint_D (N_x - M_y) = \int_0^{2\pi} \int_0^2 3r^3 \sin^2 \theta dr d\theta \\&= 8 \int_0^{2\pi} \sin^2 \theta d\theta = 8 \left[-\frac{1}{2} \sin \theta \cos \theta + \frac{\theta}{2} \right]_0^{2\pi} = 8\pi.\end{aligned}$$

Exercises

- 1 Find curvature, unit tangent and unit normal vectors of the
 - a) $\mathbf{r}(t) = 3e^{-t}(\mathbf{i} + \mathbf{j} - 2\mathbf{k})$
 - b) $\mathbf{r}(t) = e^t \sin t \mathbf{i} - \mathbf{j} + e^t \cos t \mathbf{k}$
- 2 Compute $\nabla \cdot \mathbf{F}$ and $\nabla \times \mathbf{F}$ and verify explicitly that $\nabla \cdot (\nabla \times \mathbf{F}) = 0$.
 - a) $\mathbf{F} = \sinh(xyz)\mathbf{z}$
 - b) $\mathbf{F} = 2xy\mathbf{i} + xe^y\mathbf{j} + 2z\mathbf{k}$
- 3 Compute ∇f and verify explicitly that $\nabla \times (\nabla f) = \mathbf{0}$
 - a) $f = e^{x+y+z}$
 - b) $f = \sin(xz)$
 - b) $f = xe^{-xy}$
- 4 Find the mass and center of mass of a thin, straight wire extending from the origin to $(3, 3, 3)$ if $\sigma = x + y + z$ grams per centimeter.

4. Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where
 - a) $\mathbf{F}(x, y) = e^{4y} (2x\mathbf{i} + 4x^2\mathbf{j})$, and C is the portion of the parabola $y = x^2 - 4$ from $(0, 0)$ to $(2, 0)$
 - b) $\mathbf{F}(x, y) = e^{x^2+y^2+z^2} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$ and C is the line segment from $(0, 0, 0)$ to $(1, 1, 0)$.
 - c) $\mathbf{F} = (x^2 - y)\mathbf{i} + (\cos(2y) - e^{3y} + 4x)\mathbf{j}$ and C is any square with sides of length 5.
 - d) $\mathbf{F} = e^x \cos y \mathbf{i} - e^x \sin y \mathbf{j}$ and C is any simple closed path.
6. A particle moves once counterclockwise around the circle of radius 6 about the origin, under the influence of the force $\mathbf{F}(x, y) = (e^x - y + x \cosh(x))\mathbf{i} + (y^{\frac{3}{2}} + x)\mathbf{j}$. Calculate the work done.

Representation of Surfaces in Space:

- **Explicit form** $S : z = g(x, y)$.
 - $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle$ **upward normal vector.**
 - $dS = \|\mathbf{N}\| dA = \sqrt{1 + g_x^2 + g_y^2} dx dy$
 - **Area of** $S = \iint_S dS = \iint_D \|\mathbf{N}\| dA = \iint_D \sqrt{1 + g_x^2 + g_y^2} dx dy$

For example $S : z = x^2 + y^2, 0 \leq z \leq 2$.

- $\mathbf{N} = \langle -z_x, -z_y, 1 \rangle = \langle -2x, -2y, 1 \rangle$
- $dS = \|\mathbf{N}\| dA = 2\sqrt{x^2 + y^2} dA$
- $A(S) = \iint_S dS = \iint_S 2\sqrt{x^2 + y^2} dA = 2 \int_0^{2\pi} \int_0^2 r r dr d\theta = \frac{32\pi}{3}$

- **Implicit form** $S : g(x, y, z) = 0$

For example the sphere $S : x^2 + y^2 + z^2 = 4$.

■ Parametric or vector form:

$$S : x = x(u, v), \quad y = y(u, v), \quad z = z(u, v),$$

or $S : \mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad a \leq u \leq b, \quad c \leq v \leq d$.

To parameterize a surface S , we can use

- **Cylindrical coordinates:** (r, θ, z) where
 $x = r \cos \theta, \quad y = r \sin \theta, \quad z = z, \quad r^2 = x^2 + y^2, \quad \theta = \tan^{-1} \left(\frac{y}{x} \right)$
 $u = \theta$ and $v = z$ are the parameters.
- **Spherical coordinates:** (ρ, ϕ, θ) where
 $x = \rho \sin \phi \cos \theta, \quad y = \rho \sin \phi \sin \theta, \quad z = \rho \cos \phi$
 $\rho = \sqrt{x^2 + y^2 + z^2}, \quad \phi = \cos^{-1} \left(\frac{z}{\rho} \right)$
 $u = \phi$ and $v = \theta$ are the parameters.

Example-1: Parameterize $S : x^2 + y^2 + z^2 = a^2$

- $S : \mathbf{r}(u, v) = \langle a \sin u \cos v, a \sin u \sin v, a \cos u \rangle$,
 $0 \leq u \leq \pi, 0 \leq v \leq 2\pi$.
- Normal vector: $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = a \sin u \mathbf{r}(u, v)$
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = a^2 \sin u du dv$
- $A(S) = \iint_S dS = \int_0^{2\pi} \int_0^\pi a^2 \sin u du dv = 4\pi a^2$

Example-2: Parameterize $S : x^2 + y^2 = a^2, h \leq z \leq H$.

- Vector form: $S : \mathbf{r}(u, v) = \langle a \cos u, a \sin u, v \rangle$,
 $0 \leq u \leq 2\pi, h \leq v \leq H$.
- Normal vector: $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = \langle a \cos u, a \sin u, 0 \rangle$
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = a du dv$
- $A(S) = \iint_S dS = \int_0^{2\pi} \int_h^H a du dv = 2\pi a(H - h)$

Example-3: Parameterize $S : z = \sqrt{x^2 + y^2}, h \leq z \leq H$.

- $S : \mathbf{r}(u, v) = \langle v \cos u, v \sin u, v \rangle, (u, v) \in [0, 2\pi] \times [h, H]$
- Normal vector: $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = \langle v \cos u, v \sin u, -v \rangle$
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = v\sqrt{2} du dv$
- $A(S) = \iint_S dS = \int_0^{2\pi} \int_h^H v\sqrt{2} du dv = \sqrt{2}\pi(H^2 - h^2)$
- A surface $S : \mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$, is **smooth** if $\mathbf{r}_u$ and $\mathbf{r}_v$ are continuous and is never zero on the parameter domain, $D = \{(u, v) | a \leq u \leq b, c \leq v \leq d\}$.

Example-3: Parameterize $S : z = \sqrt{x^2 + y^2}$, $h \leq z \leq H$.

- $S : \mathbf{r}(u, v) = \langle v \cos u, v \sin u, v \rangle$, $(u, v) \in [0, 2\pi] \times [h, H]$
- Normal vector: $\mathbf{N} = \mathbf{r}_u \times \mathbf{r}_v = \langle v \cos u, v \sin u, -v \rangle$
- $dS = \|\mathbf{r}_u \times \mathbf{r}_v\| du dv = v\sqrt{2} du dv$
- $A(S) = \iint_S dS = \int_0^{2\pi} \int_h^H v\sqrt{2} du dv = \sqrt{2}\pi(H^2 - h^2)$

- A surface $S : \mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle$, is **smooth** if $\mathbf{r}_u$ and $\mathbf{r}_v$ are continuous and is never zero on the parameter domain, $D = \{(u, v) \mid a \leq u \leq b, c \leq v \leq d\}$.
- A surface S is orientable (or two-sided) if it is possible to define a unit normal vector $\mathbf{n}$ at each point (x, y, z) not on the boundary of the surface and if $\mathbf{n}$ is a continuous function of (x, y, z) .

Surface Integral of a Scalar-Field

Definition

Let $S : \mathbf{r}(u, v) = x(u, v)\mathbf{i} + y(u, v)\mathbf{j} + z(u, v)\mathbf{k}$, $a \leq u \leq b$, $c \leq v \leq d$ be a smooth surface and f is a continuous function defined on S , then the integral of f over S is called **surface integral** and defined as

$$\begin{aligned} \iint_S f dS &= \iint_D f(\mathbf{r}(u, v)) \|\mathbf{N}\| du dv \\ &= \int_a^b \int_c^d f(x(u, v), y(u, v), z(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| du dv \end{aligned}$$

- If $S : z = g(x, y)$, $(x, y) \in D$, then

$$\iint_S f dS = \iint_D f(x, y, g(x, y)) \sqrt{1 + g_x^2 + g_y^2} dA$$

where D is the projection of S on the xy -plane.

Examples: Evaluate $\iint_S f dS$, where

- 1 $f = 3z$ and S is the portion of the plane $x + y + z = 2$ inside the cylinder $x^2 + y^2 = 1$.
- 2 $f = x^2 + y^2 + z^2$ and S is the part of the cylinder $x^2 + y^2 = 9$ between the planes $z = 0$ and $z = 2$.

Solutions:

- 1 $S: x + y + z = 2, \quad D = \{(x, y) | x^2 + y^2 \leq 1\}$ and $\mathbf{N} = \langle 1, 1, 1 \rangle$.

$$\iint_S 3z dS = 3\sqrt{3} \int_0^{2\pi} \int_0^1 [2 - r \cos \theta - r \sin \theta] r dr d\theta = 6\pi.$$
- 2 $S: \mathbf{r}(u, v) = \langle 3 \cos u, 3 \sin u, v \rangle, \quad D = [0, 3] \times [0, 2\pi]$
 $f = 9 + v^2, \quad \mathbf{N} = \langle 3 \cos u, 3 \sin u, 0 \rangle$ and $\|\mathbf{N}\| = 3$

$$\iint_S (x^2 + y^2 + z^2) dS = \int_0^{2\pi} \int_0^2 (27 + 3v^2) dv du = 124\pi$$

- If a thin sheet (say, of aluminum foil) has the shape of a surface S and the density (mass per unit area) at the point (x, y, z) is $\sigma(x, y, z)$, then the total mass of the sheet is

$$m = \iint_S \sigma dS$$

and the center of mass is $(\bar{x}, \bar{y}, \bar{z})$ where

$$\bar{x} = \frac{1}{m} \iint_S x \sigma dS, \quad \bar{y} = \frac{1}{m} \iint_S y \sigma dS, \quad \bar{z} = \frac{1}{m} \iint_S z \sigma dS.$$

Example:

Find the mass and center of mass of a thin funnel in the shape of a cone $z = \sqrt{x^2 + y^2}$, $1 \leq z \leq 4$ if its density function is $\sigma(x, y, z) = 10 - z$.

Solution:

- $S : \mathbf{r}(\mathbf{u}, \mathbf{v}) = \langle v \cos u, v \sin u, v \rangle, \quad D = [0, 2\pi] \times [1, 4]$
- $\mathbf{N} = \langle v \cos u, v \sin u, -v \rangle$ and $||\mathbf{N}|| = \sqrt{2}v$.
- $\sigma = 10 - z = 10 - v$
- $m = \iint_S \sigma dS = \iint_D (10 - v)v\sqrt{2} \, dudv = \sqrt{2} \int_0^{2\pi} \int_1^4 (10v - v^2) \, dvdu = 118\sqrt{2}\pi$
- Let $(\bar{x}, \bar{y}, \bar{z})$ be the center of mass the funnel, where
 - $\bar{x} = \frac{1}{m} \iint_S x \sigma dS = \frac{1}{118\pi\sqrt{2}} \int_0^{2\pi} \int_1^4 (10 - v)v \cos u \sqrt{2} v \, dvdu = 0$
 - $\bar{y} = \frac{1}{m} \iint_S y \sigma dS = \frac{1}{118\pi\sqrt{2}} \int_0^{2\pi} \int_1^4 (10 - v)v \sin u \sqrt{2} v \, dvdu = 0$
 - $\bar{z} = \frac{1}{m} \iint_S z \sigma \sqrt{2} dS = \frac{1}{118\pi\sqrt{2}} \int_0^{2\pi} \int_1^4 (10 - v)\sqrt{2} v \, dvdu = \frac{17}{59}$

Surface Integral of Vector Fields

Definition

- Let $\mathbf{F}$ be a continuous vector field defined on an oriented surface $S : \mathbf{r} = \mathbf{r}(u, v)$, $a \leq u \leq b$, $c \leq v \leq d$ with unit normal vector $\mathbf{n}$. Then the surface integral of $\mathbf{F}$ over S is

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \int_a^b \int_c^d \mathbf{F}(\mathbf{r}(u, v)) \cdot \mathbf{N} \, dv \, du$$

where $\mathbf{N}$ is a normal vector in the direction of $\mathbf{n}$.

- For $S : z = g(x, y)$, $(x, y) \in D$ and $\mathbf{N} = \langle -g_x, -g_y, 1 \rangle$,

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iint_D \mathbf{F}(x, y, g(x, y)) \cdot \langle -g_x, -g_y, 1 \rangle \, dA$$

- $\iint_S \mathbf{F} \cdot \mathbf{n} dS$ is also called the **flux** of $\mathbf{F}$ across S .

Example-1:

Find the flux of $\mathbf{F} = x\mathbf{i} + y\mathbf{j} - z\mathbf{k}$ across the part of the plane $x + 2y + z = 4$ in the first octant.

Solution

$$\blacksquare S: z = 4 - x - 2y, \quad D: 0 \leq y \leq 2, \quad 0 \leq x \leq 4 - 2y.$$

$$\blacksquare \mathbf{N} = \langle -z_x, -z_y, 1 \rangle = \langle 1, 2, 1 \rangle.$$

$$\blacksquare \mathbf{F} \cdot \mathbf{N} = x + 2y - z = 2x + 4y - 4$$

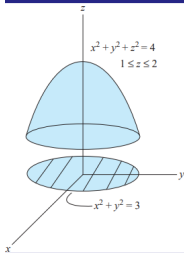
$$\begin{aligned} \text{flux} &= \iint_S \mathbf{F} \cdot \mathbf{n} dS = \int_0^2 \int_0^{4-2y} (2x + 4y - 4) dx dy \\ &= \int_0^2 (x^2 + 4xy - 4x) \Big|_0^{4-2y} dy = \int_0^2 (8y - 4y^2) dy = \frac{16}{3}. \end{aligned}$$



Example-2:

Find the flux of $\mathbf{F} = xz\mathbf{i} - y\mathbf{k}$ across the part of the sphere $x^2 + y^2 + z^2 = 4$ above the plane $z = 1$.

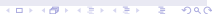
Solution:



- $\blacksquare$ The plane $z = 1$ intersects the sphere in the circle $x^2 + y^2 = 3$. This circle projects onto the circle $x^2 + y^2 = 3$ in the xy -plane.
- $\blacksquare S: x^2 + y^2 + z^2 = 4, \quad 1 \leq z \leq 2$

$$\phi = \cos^{-1} \left(\frac{z}{\rho} \right) = \cos^{-1} \left(\frac{1}{2} \right) = \frac{\pi}{3}.$$
- $\blacksquare 0 \leq \phi \leq \frac{\pi}{3}, \quad 0 \leq \theta \leq 2\pi.$

$$D = \{(u, v) \mid 0 \leq u \leq \frac{\pi}{3}, \quad 0 \leq v \leq 2\pi\}$$



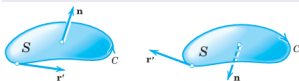
- $S: \mathbf{r}(u, v) = \langle 2 \sin u \cos v, 2 \sin u \sin v, 2 \cos u \rangle$.
- $\mathbf{N} = 2 \sin u \mathbf{r}'(u, v) = \langle 4 \sin^2 u \cos v, 4 \sin^2 u \sin v, 4 \sin u \cos u \rangle$
- $\mathbf{F}(\mathbf{r}(u, v)) = \langle 4 \sin u \cos u \cos v, 0, -2 \sin u \sin v \rangle$
- $\mathbf{F} \cdot \mathbf{N} = 16 \sin^3 u \cos u \cos^2 v - 8 \sin^3 u \cos u \sin v$
 $= 8 \sin^3 u \cos u (1 + \cos 2v - \sin v)$

$$\begin{aligned}
 flux &= \iint_S \mathbf{F} \cdot \mathbf{n} dS \\
 &= 8 \int_0^{\pi/3} \sin^3 u \cos u \left[\int_0^{2\pi} (1 + \cos 2v - \sin v) dv \right] du \\
 &= 16\pi \int_0^{\pi/3} \sin^3 u \cos u du \\
 &= 9\pi
 \end{aligned}$$

Stokes' and Divergent Theorems

Stokes' Theorem

Let S be a piecewise smooth oriented surface in space and let the boundary of S be a piecewise smooth simple closed curve C . Let $\mathbf{F}$ be a continuous vector function that has continuous first partial derivatives in a domain in space containing S .



$$\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$$

or

$$\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \int_C \mathbf{F} \cdot d\mathbf{r}$$

Example-1: Verify that Stokes' Theorem

$\mathbf{F} = \langle y^2, x, z^2 \rangle$ and S is the part of the paraboloid $z = x^2 + y^2$ that lies below the plane $z = 1$, oriented upward.

Solution:

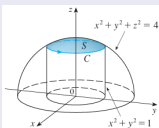
- $C : x^2 + y^2 = 1, z = 1 \Rightarrow \mathbf{r}(t) = \langle \cos t, \sin t, 1 \rangle, 0 \leq t \leq 2\pi.$
 - $\mathbf{r}'(t) = \langle -\sin t, \cos t, 0 \rangle$ and $\mathbf{F}(\mathbf{r}(t)) = \langle \sin^2 t, \cos t, 1 \rangle$
 - $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = \cos^2 t - \sin^3 t = \frac{1+\cos 2t}{2} - \sin t + \cos^2 t \sin t$
 - $\int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \left(\frac{1+\cos 2t}{2} - \sin t + \cos^2 t \sin t \right) dt = \pi.$
- Surface integral: $S : z = x^2 + y^2, D : x^2 + y^2 \leq 1.$
- $\text{curl } \mathbf{F} = \langle P_y - N_z, M_z - P_x, N_x - M_y \rangle = \langle 0, 0, 1 - 2y \rangle$
 - $\mathbf{N} = \langle -2x, -2y, 1 \rangle$ and $\text{curl } \mathbf{F} \cdot \mathbf{N} = 1 - 2y = 1 - 2r \sin \theta$
 - $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \int_0^{2\pi} \int_0^1 (1 - 2r \sin \theta) r dr d\theta = \pi$

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Example-2:

Use Stokes' Theorem to compute the integral $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS$, where $\mathbf{F} = \langle z^2, xz^2, x^2y^2 \rangle$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.

Solution:



- C is the intersection of the surfaces: $x^2 + y^2 + z^2 = 4$ and $x^2 + y^2 = 1.$
- $C : \mathbf{r}(t) = \langle \cos t, \sin t, \sqrt{3} \rangle, 0 \leq t \leq 2\pi.$
- $\mathbf{r}'(t) = \langle -\sin t, \cos t, 0 \rangle$
- $\mathbf{F}(\mathbf{r}(t)) = \langle 3, 3 \cos t, \sin^2 t \cos^2 t \rangle$
- $\mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) = (3 \cos^2 t - 3 \sin t) = \frac{3+3 \cos 2t}{2} - 3 \sin t$
- $\iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} \left(\frac{3+3 \cos 2t}{2} - 3 \sin t \right) dt = 3\pi.$

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Example-3: Use Stokes' Theorem to evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$, where

$\mathbf{F} = \langle xy, 2z, 3y \rangle$ and C is the curve of intersection of the plane $x + z = 5$ and the cylinder $x^2 + y^2 = 9$.

Solution

- $S : x + z = 5, \quad D : x^2 + y^2 \leq 9 = [0, 3] \times [0, 2\pi]$.
- $\mathbf{N} = \langle 1, 0, 1 \rangle$
- $\text{curl } \mathbf{F} = \langle 1, 0, -x \rangle$
- $\text{curl } \mathbf{F} \cdot \mathbf{N} = 1 - x = 1 - r \cos \theta$
- $\int_C \mathbf{F} \cdot d\mathbf{r} = \iint_S \text{curl } \mathbf{F} \cdot \mathbf{n} dS = \int_0^{2\pi} \int_0^3 (1 - r \cos \theta) r \, dr d\theta = 9\pi$.

Divergence Theorem:

Theorem: Divergent Theorem

Let T be a closed bounded region in space whose boundary is a piecewise smooth orientable surface S . Let $\mathbf{F} = \langle M, N, P \rangle$ be a vector field that is continuous and has continuous first partial derivatives in some domain containing T . Then

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \text{div} \mathbf{F} dV = \iiint_T (M_x + N_y + P_z) dV$$

($\mathbf{n}$ is the outward unit normal vector of S .)

Example-1:

Evaluate $\iint_S (7x\mathbf{i} - z\mathbf{k}) \cdot \mathbf{n} dS, \quad S : x^2 + y^2 + z^2 = 4$

$$\blacksquare \operatorname{div} \mathbf{F} = 7 - 1 = 6$$

$$\blacksquare \iint_S (7x\mathbf{i} - z\mathbf{k}) \cdot \mathbf{n} dS = \iiint_T \operatorname{div} \mathbf{F} dV = \iiint_T 6 dV = 6V(T) = 64\pi$$

Example-2: Use the Divergence Theorem to calculate $\iint_S \mathbf{F} \cdot \mathbf{n} dS$, where

- a) $\mathbf{F} = \langle x^3, ze^x, 3zy^2 \rangle$ and S is the surface of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = -1$ and $z = 2$.
- b) $\mathbf{F} = \langle \cos z + xy^2, xe^{-z}, \sin y + x^2z \rangle$ and S is the surface of the solid bounded by the paraboloid $z = x^2 + y^2$ and the plane $z = 4$.
- c) $\mathbf{F} = \langle x^4, -x^3z^2, 4xy^2z \rangle$ and S is the surface of the solid bounded by the cylinder $x^2 + y^2 = 1$ and the planes $z = x + 2$ and $z = 0$.

Solutions:

a) $\operatorname{div} \mathbf{F} = 3x^2 + 3y^2 = 3r^2, \quad T = [0, 1] \times [0, 2\pi] \times [-1, 2]$

$$\iint_S \mathbf{F} \cdot \mathbf{n} dS = \iiint_T \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^1 \int_{-1}^2 3r^3 dz dr d\theta = \frac{9\pi}{2}$$

b) $\operatorname{div} \mathbf{F} = y^2 + x^2 = r^2, \quad T = [0, 2] \times [0, 2\pi] \times [r^2, 4]$

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_T \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^2 \int_{r^2}^4 r^3 dz dr d\theta \\ &= \int_0^{2\pi} \int_0^2 (4r^3 - r^5) dr d\theta = \frac{32\pi}{3} \end{aligned}$$

c) $\operatorname{div} \mathbf{F} = 4x^3 + 4xY^2 = 4x(x^2 + y^2) = 4r^3 \cos \theta$

$$T = \{(r, \theta, z) \mid 0 \leq \theta \leq 2\pi, \quad 0 \leq r \leq 1, \quad 0 \leq z \leq 2 + r \cos \theta\}$$

$$\begin{aligned} \iint_S \mathbf{F} \cdot \mathbf{n} dS &= \iiint_T \operatorname{div} \mathbf{F} dV = \int_0^{2\pi} \int_0^1 \int_0^{2+r \cos \theta} 4r^4 \cos \theta dz dr d\theta \\ &= \frac{2\pi}{5} \end{aligned}$$

Exercise

- For the curve given by $\mathbf{r}(t) = \langle \frac{1}{3}t^3 + t^2, 2t \rangle$, find
 - the length
 - the unit tangent vector
 - the unit normal vector
 - the curvature
- Evaluate the line integral
 - $\int_C yz \cos x \, ds$, $C: x = t, y = 3 \cos t, z = 3 \sin t, 0 \leq t \leq \pi$.
 - $\int_C x e^{yz} \, ds$, C is the line segment from $(0, 0, 0)$ to $(1, 2, 3)$.
 - $\int_C (2x + 9z) \, ds$, $C: x = t, y = t^2, z = t^3, 0 \leq t \leq 1$.
 - $\int_C (x - y) \, ds$, C is the portion of $y = x^3$ from $(1, 1)$ to $(-1, -1)$
 - $\int_C (x^2 + y^2) \, ds$, C is the half-circle $x^2 + y^2 = 16$ from $(4, 0)$ to $(-4, 0)$, $y \geq 0$.
- Find the mass of a thin wire lying along the curve $\mathbf{r}(t) = \sqrt{2}t\mathbf{i} + \sqrt{2}t\mathbf{j} + (4 - t^2)\mathbf{k}$, $0 \leq t \leq 1$ with density $\sigma(t) = 3t$

- Evaluate the line integral
 - $\int_C 3x^2y^2 \, dx + (2x^3y - 4) \, dy$, where C runs from $(1, 2)$ to $(-1, 1)$.
 - $\int_C 3(2x \cos z - x^2) \, dx + (z - 2y) \, dy + (y - x^2 \sin z) \, dz$, where C runs from $(3, -2, 0)$ to $(1, 0, \pi)$.
 - $\int_C \mathbf{F} \cdot d\mathbf{r}$, $\mathbf{F} = \left\langle \frac{x}{1+x^2} - y, 3x - \tan \frac{y}{2} \right\rangle$, C is the portion of $y = x^2$ from $(-1, 1)$ to $(1, 1)$, followed by the portion of $y = 2 - x^2$ from $(1, 1)$ to $(-1, 1)$.
 - $\int_C (\sqrt{x^2 + 1} - x^2y) \, dx + (xy^2 - y^{5/3}) \, dy$, C is the circle $x^2 + y^2 = 4$ oriented clockwise
 - $\int_C \mathbf{F} \cdot d\mathbf{r}$, $\mathbf{F} = xy\mathbf{i} + (xy^2 - e^{\cos y})\mathbf{j}$ and C is the triangle with vertices $(0, 0)$, $(3, 0)$ and $(0, 5)$.
 - $\int_C \mathbf{F} \cdot d\mathbf{r}$, $\mathbf{F} = 2x^2\mathbf{i} + 4y^2\mathbf{j} + e^{8z^2}\mathbf{k}$ and C is the intersection of $z = x^2 + y^2$ and $z = 8 - y$, oriented counterclockwise.
 - $\int_C \mathbf{F} \cdot d\mathbf{r}$, $\mathbf{F} = \langle yz, 2xz, e^{xy} \rangle$ and C is the circle $x^2 + y^2 = 16$, $z = 5$, oriented counterclockwise.
 - $\int_C \mathbf{F} \cdot d\mathbf{r}$, $\mathbf{F} = \langle x + y^2, y + z^2, x^2 + z \rangle$ and C is the triangle with vertices $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$, oriented counterclockwise.

5. Evaluate the surface integral:

- $\iint_S xy \, dS$, S is the triangular region with vertices $(1, 0, 0)$, $(0, 2, 0)$ and $(0, 0, 2)$.
- $\iint_S yz \, dS$, S is the surface with parametric equations $x = u^2$, $y = u \sin v$, $z = u \cos v$, $0 \leq u \leq 1$, $0 \leq v \leq \frac{\pi}{2}$.
- $\iint_S z \, dS$, S is the part of the paraboloid $z = x^2 + y^2$ that lies inside the cylinder $x^2 + y^2 = 4$.
- $\iint_S z \, dS$, S is the part of the sphere $x^2 + y^2 + z^2 = 4$ that lies inside the cylinder $x^2 + y^2 = 1$ and above the xy -plane.
- $\iint_S z \, dS$, S is the part of the cylinder $x^2 + y^2 = 1$ that lies between the planes $z = 0$ and $z = 3$ in the first octant.
- $\iint_S (x^2z + y^2z) \, dS$, S is the part of the plane $z = 4 + x + y$ that lies inside the cylinder $x^2 + y^2 = 4$.

6. Find the mass and center of mass of the object corresponding to the portion of the paraboloid $z = x^2 + y^2$ inside the cylinder $x^2 + y^2 = 4$ and mass density $\rho(x, y, z) = z$.

7. Evaluate the surface integral :

- $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, $\mathbf{F} = x\mathbf{i} + y\mathbf{j} + z^4\mathbf{k}$ and S is the part of the cone $z = \sqrt{x^2 + y^2}$ beneath the plane $z = 1$ ($\mathbf{n}$ downward).
- $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, $\mathbf{F} = x\mathbf{i} - z\mathbf{j} + y\mathbf{k}$ and S is the part of the sphere $x^2 + y^2 + z^2 = 4$ in the first octant ($\mathbf{n}$ toward the origin).
- $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, $\mathbf{F} = xy^2\mathbf{i} + x^2y\mathbf{j} + y\mathbf{k}$ and S is the surface of the region enclosed by the cylinder $x^2 + y^2 = 1$ and the planes $z = -1$ and $z = 1$, ($\mathbf{n}$ outward).
- $\iint_S \mathbf{F} \cdot \mathbf{n} \, dS$, $\mathbf{F} = 3xz^2\mathbf{i} + y^3\mathbf{j} + 3x^2z\mathbf{k}$ and S is the surface of the solid region enclosed by the cone $z = \sqrt{x^2 + y^2}$ and the hemisphere $z = \sqrt{8 - x^2 - y^2}$, ($\mathbf{n}$ outward).
- $\iint_S \text{curl} \mathbf{F} \cdot \mathbf{n} \, dS$, $\mathbf{F} = \langle x^2z, ze^{xy^2} - x, x \ln y^2 \rangle$ and S is the portion of the paraboloid $z = 1 - x^2 - y^2$ above the xy -plane, ($\mathbf{n}$ upward).
- $\iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} \, dS$, $\mathbf{F} = \langle x^2 + y^2, ze^{x^2+y^2}, e^{x^2+z^2} \rangle$ and S is the portion of the cone $z = \sqrt{x^2 + y^2}$ below the sphere $x^2 + y^2 + z^2 = 2$ ($\mathbf{n}$ downward).